NANDYALA GIRIDHAR

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SUMMARY

Agentic AI Engineer building end-to-end multi-agent systems using LangGraph, CrewAI, FastAPI, and Docker. Built an NL-to-SQL agent that reached 94% query accuracy at 1.6s median latency, reduced execution failures from 32% to 3% with self-correction loops, and blocked 18/18 destructive queries through human-in-the-loop review. Also built multi-agent research, RAG, and multimodal AI systems.

TECHNICAL SKILLS

Agentic AI:	LangGraph, CrewAI, multi-agent workflows, tool/function calling, human-in-the-loop, self-correction loops, guardrails
LLMs & RAG:	OpenAI API, Google Gemini API, Groq, Ollama, LangChain, RAG pipelines, vector databases (Qdrant), embeddings, semantic search, prompt engineering
Production & Deployment:	FastAPI, Docker, Render, REST APIs, Git/GitHub, n8n, Streamlit
Evaluation & Observability:	agent evaluation, LLM evaluation, execution-accuracy benchmarking, safety testing, LangSmith tracing
Computer Vision:	YOLOv8, ByteTrack, EasyOCR, OpenCV
Languages & Data:	Python, SQL, JSON, Pandas

PROJECTS

Enterprise AI Data Copilot: Production Agentic NL-to-SQL System [GitHub] [Live App]

Python | LangGraph | FastAPI | Docker | LangSmith | Render | SQL | Streamlit

- Designed a multi-agent LangGraph workflow (schema reflection, SQL generation, validation, execution, reporting) that reached 94% text-to-SQL accuracy on multi-table join benchmarks.
- Implemented self-correction retry loops that return database errors to the agent for automated query repair, cutting execution failures from 32% to 3% across 120+ schema test queries.
- Enforced human-in-the-loop review and safety validation, blocking 18/18 destructive test queries (DROP, DELETE, TRUNCATE) before database execution.
- Instrumented every agent run with LangSmith tracing to inspect tool calls, latency, and cost, achieving a 1.6s median query response time in benchmark testing.
- Shipped the agent as a versioned FastAPI service (/api/v1/query), containerized with Docker and deployed on Render, with a Streamlit front end for charts and automated analytical reports.

Multi-Agent AI Job Market & Resume Researcher [GitHub] [Live App]

Python | CrewAI | Gemini API | Streamlit

- Built a CrewAI multi-agent research workflow using Gemini API to analyze job-market demand and generate structured resume recommendations.
- Delivered the system as a deployed Streamlit application with a public repository and live demo.

Traffic Flow & Speed Analytics: Multimodal AI System [GitHub] [Live App]

YOLOv8 | ByteTrack | EasyOCR | OpenCV | Streamlit | Gemini API

- Engineered a real-time vision pipeline for vehicle detection, tracking, speed estimation, and license plate recognition, processing video at 25 FPS with 89% overall detection accuracy.
- Integrated the Gemini API to generate automated safety risk assessments and traffic insights from analyzed footage, surfaced in a Streamlit dashboard with speed-violation detection and e-challan snapshots.

PDF Knowledge Assistant: RAG System [GitHub] [Live App]

Python | LangChain | Qdrant Cloud | Gemini API | Streamlit

- Built a RAG pipeline that chunks multi-page PDFs, embeds them into Qdrant Cloud, and retrieves passages by semantic similarity.
- Grounded Gemini responses in the retrieved passages to answer questions directly from document content.

EXPERIENCE

AI / Generative AI Project Developer — **Independent Projects | 2025-2026**

- Developed and deployed AI applications using LangGraph, CrewAI, RAG, and computer vision, covering end-to-end development from prototyping to deployment.
- Built production-style AI workflows with FastAPI, Docker, and Streamlit, integrating evaluation, error handling, and safety validation across projects.

EDUCATION

Centurion University of Technology and Management — **2022-2026**

Bachelor of Technology in Computer Science and Engineering | CGPA: 8.20

CERTIFICATIONS

- Agentic AI Certified Foundations Associate, Oracle University
- Oracle Cloud Infrastructure (OCI) AI Database Certified Associate, Oracle University